

Multiple Choice

1. **Answer:** A

Options: A) No, an attacker can simply guess the tag for messages; B) It depends on the details of the MAC; C) Yes, the attacker cannot generate a valid tag for any message; D) Yes, the PRG is pseudorandom

Note: Correct option: A - No, an attacker can simply guess the tag for messages

2. **Answer:** A

Options: A) Message Nonrepudiation; B) Message Integrity; C) Message Confidentiality; D) Message Sending

Note: Correct option: A - Message Nonrepudiation

3. **Answer:** A

Options: A) SYN stealth scan; B) TCP connect; C) XMAS tree scan; D) ACK scan

Note: Correct option: A - SYN stealth scan

4. **Answer:** C

Options: A) Active, inactive, standby; B) Open, half-open, closed; C) Open, filtered, unfiltered; D) Active, closed, unused

Note: Correct option: C - Open, filtered, unfiltered

5. **Answer:** A

Options: A) Freenet; B) ARPANET; C) Stuxnet; D) Internet

Note: Correct option: A - Freenet

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'WPA 3'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Confidentiality, Integrity, Non repudiation and Authentication'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Message Integrity'.

Long Form Response

11. **Answer:** `def perimeter_pentagon(a): | return perimeter`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def get_pairs_count(arr, n, sum): | return count`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key